Chapter 8: Project Quality Management

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations

Twittie Senivongse (ed.)

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Learning Objectives (1 of 2)

- Define project quality management and understand how quality relates to various aspects of IT projects
- Describe quality management planning and how quality and scope management are related
- Discuss the importance of managing quality and quality assurance
- Explain the main outputs of the quality control process

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Learning Objectives (2 of 2)

- List and describe the tools and techniques for quality control, such as the Basic Tools of Quality, statistical sampling, Six Sigma, and testing
- Describe modern quality management
- Describe how leadership, the cost of quality, organizational influences, expectations, cultural differences, and maturity models relate to improving quality in IT projects
- Discuss how software can assist in project quality management
- Discuss considerations for agile/adaptive environments

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What Is Quality?

- International Organization for Standardization (ISO) definition of quality
  - "Totality of characteristics of an entity that bear on its ability to satisfy stated or implied needs" (ISO8402:1994)
  - "The degree to which a set of inherent characteristics fulfils requirements" (ISO9000:2000)
- Other definitions of quality
  - Conformance to requirements
    - Project's processes and products meet written specifications
  - Fitness for use
    - Product can be used as it was intended

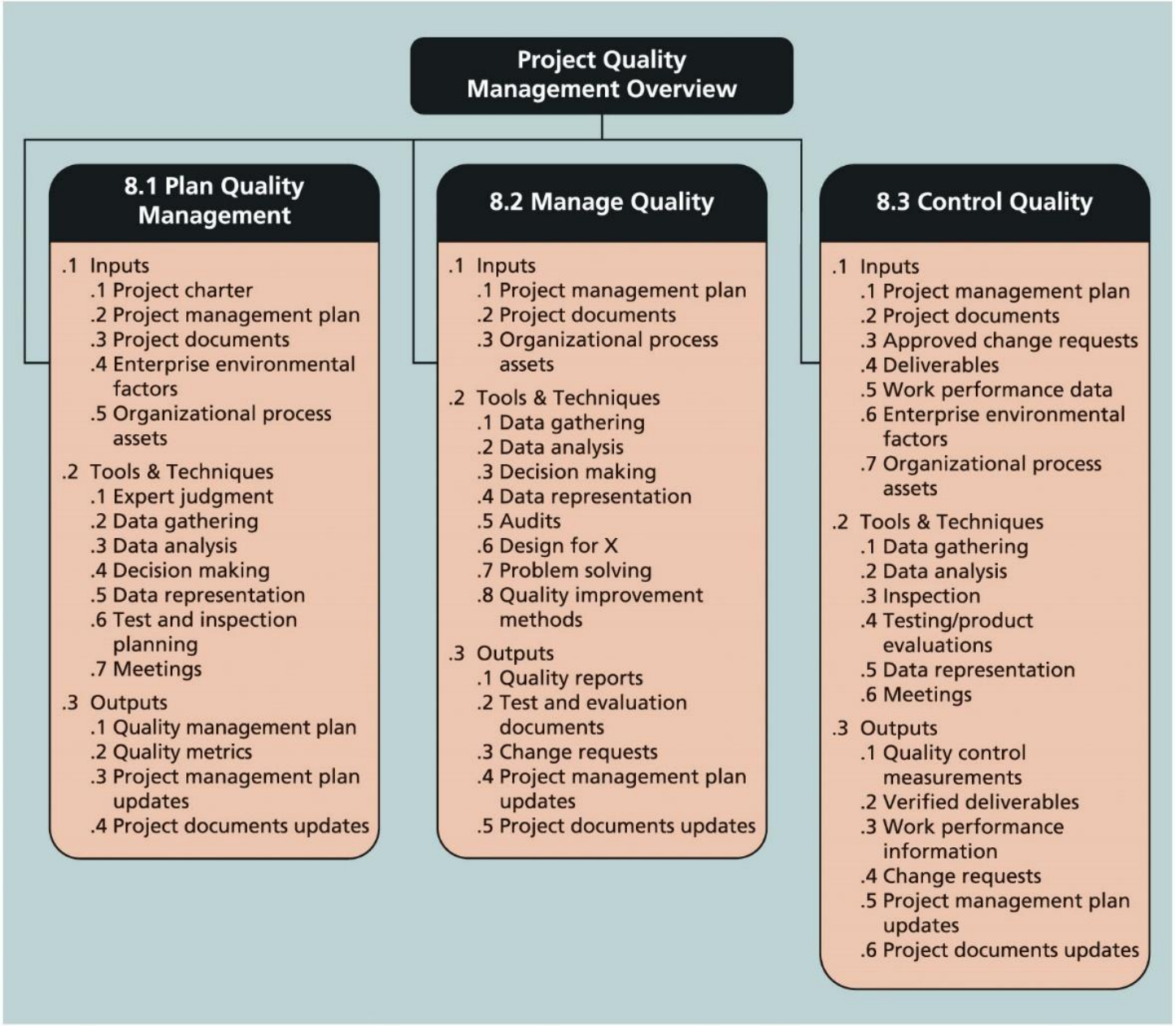
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Project Quality Management

- Project quality management ensures the project will satisfy the needs for which it was undertaken
- Project quality management processes

1. Planning quality management
   - Identifying which quality standards are relevant to the project, and how to satisfy them, e.g.
     - Allowing system growth,
     - Reasonable response time
     - Consistent and accurate information
     - Guaranteed service time
   - Identifying metrics as the standards of measurement, e.g.
     - Failure rate
     - Availability
     - Customer rating
2. Managing quality
   - Translating the quality management plan into executable quality activities
3. Controlling quality
   - Monitoring specific project results to ensure they comply with the relevant quality standards

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FIGURE 8-1 Project quality management overview

1. Planning Quality Management (1 of 3)

- The current thrust in modern quality management is the prevention of defects through
  - Selecting proper materials
  - Training and indoctrinating people in quality
  - Planning a process that ensures the appropriate outcome
- Project quality management planning
  - Identifies quality standards for each unique project
  - Designs quality into the products of the project
  - Designs quality into the processes involved in managing the project

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Planning Quality Management (2 of 3)

- Aspects of project scope that affect quality
  - Functionality: degree to which a system performs its intended function
    - E.g. ability of the EIS to track product sales by product group, country, salesperson
  - Features: system's special characteristics that appeal to users
    - E.g. graphical user interface with icons, menus, online help
  - System outputs: screens and reports the system generates
  - Performance: how well a product or service performs the customer's intended use
  - Reliability: ability of a product or service to perform as expected under normal conditions
  - Maintainability: ease of performing maintenance on a product
  - Etc.
- All project stakeholders must work together to balance the quality, scope, time, and cost dimensions of the project

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Planning Quality Management (3 of 3)

- A few quality standards
  - ISO 9000: International standards on quality management and quality assurance (https://asq.org/quality-resources/iso-9000)
    - Three-part, continuous cycle of planning, controlling, and documenting quality in an organization
  - ISO/IEC 25010: Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models
    (https://www.iso.org/obp/ui/#iso:std:iso-iec:25010:ed-1:en)
    (https://iso25000.com/index.php/en/iso-25000-standards/iso-25010)
  - IEEE 730: IEEE Standard for Software Quality Assurance Processes (https://ieeexplore.ieee.org/document/6835311)
    (https://en.wikipedia.org/wiki/Software_quality_assurance)

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2. Managing Quality (1 of 7)

- How to ensure delivery of high-quality products or services
- Includes all quality assurance activities, product design, and process improvements
  - Quality assurance includes all the activities related to satisfying the relevant quality standards for a project
  - Another goal is continuous quality improvement

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Managing Quality (2 of 7)

- Lean
  - Maximize customer value while minimizing waste
  - Created at Toyota but owing its origin to Ford Company
  - Identify wasteful activities and the ways these wastes are killing business efficiency, and eliminate or reduce these wasteful activities



What is lean production https://www.youtube.com/watch?v=J4v-HjY3R0Y
An Introduction to Lean Thinking https://www.youtube.com/watch?v=RxDw0Q_gVt0

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Image from https://blog.triaster.co.uk/blog/what-are-the-7-wastes-killing-business

Managing Quality (3 of 7)

- Lean (cont.)
  - Waste is any non-value added activity, adds problems, and blocks the flow through the process.

Information waste, e.g.
- Redundant input and output data
- Incompatible information systems
- Manual checking of data that has been entered electronically
- Data input is never used
- Reentering data
- Converting format (e.g.copy content in excel file into word)
- Unnecessary data
- Unavailable data
- Missing or unknown data
- Unclear or incorrect data
- Data discrepancies

Process waste, e.g.
- Defects
- Rework
- Workarounds
- Variable flows (times when there is too much work, and times when there is not enough)
- Too much inventory
- Incompatible work
- Overproduction
- Waiting
- Overprocessing (e.g. use a tool that is unnecessarily too expensive or sophisticated)

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Managing Quality (4 of 7)

- Lean (cont.)
  - Waste is any non-value added activity, adds problems, and blocks the flow through the process.

Physical environment waste (inefficient movement of people and objects), e.g
- Traveling to another office location for a meeting
- Organizing a messy conference room before a meeting
- Going to your admin office for a replacement pen

People waste, e.g.
- Unclear role (unclear responsibility, authority, and accountability)
- Lack of training
- Task interruptions
- Multitasking
- Underutilization of talent
- Hierarchy of structure (that blocks the flow of ideas, work documents, and decisions)
- Recruitment errors (hiring mistakes)
- Lack of strategic focus
- Handoffs (when work is passed from one person to another. Often handoffs require mental reshuffling of work. Eliminate the handoff and finish the work.)

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Managing Quality (5 of 7)

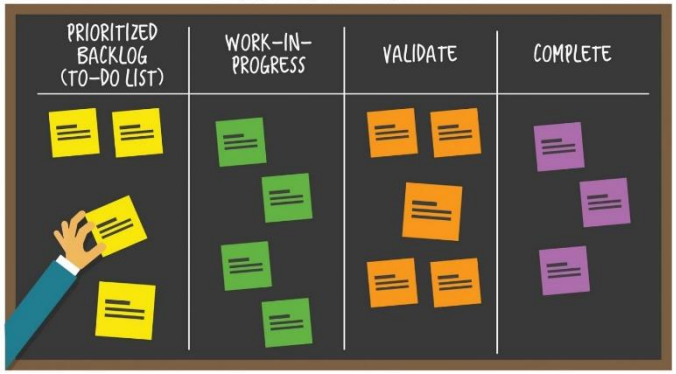
- Kaizen
  - Japanese word for improvement or change for the better 改善
  - First practiced in Japanese businesses, spearheaded by Toyota, after World War II
  - Influenced in part by American business and quality management experts
  - A mindset or philosophy, rather than a tool to use
    - Improving the business as a whole by creating a standard way of working, increasing efficiency, and eliminating business waste
    - Collecting knowledge from all employees within an organization to accomplish incremental improvements on a regular basis; achievement is not on any individual but collated as a whole

Toyota Kaizen https://www.youtube.com/watch?v=wot9DFzFRLU

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Managing Quality (6 of 7)

- Kanban
  - Japanese term for signboard or billboard 看板
  - Developed in Japan by Toyota as a scheduling system for lean manufacturing and just-in-time manufacturing (visual workflow)
  - Uses visual cues for managing work as it moves through a process, emphasizing continuous flow of work
  - In agile software development, team can
    - Place Kanban cards on the board to show status of work in backlog such as to-do, in progress, validate, and complete
    - Impose work-in-progress (WIP) limit in any one workflow step at a time, e.g. development WIP limit = 3, testing WIP limit = 2
    - If development already contains 3 items, developers should not pull another item into development. Instead, they should help move existing work forward. This prevents too much unfinished work from accumulating.
    - Continuous process as there are no sprints
  - Kanban board can be used with Scrum



Scrum vs Kanban https://www.youtube.com/watch?v=rIaz-l1Kf8w

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Managing Quality (7 of 7)

- Benchmarking
  - Comparing specific project practices or product characteristics to those of other projects or products within or outside the performing organization
- Quality audit
  - Structured review of specific quality management activities
  - Helps identify lessons learned that could improve performance on current or future projects
  - Performed by in-house auditors or third parties, as scheduled or at random
  - Involving designing, applying, and analyzing quality metrics

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3. Controlling Quality

- Main outputs of quality control
  - Acceptance decisions
    - Determine if the products or services produced as part of the project will be accepted or rejected
    - Accepted: validated deliverables
    - Rejected: rework
  - Rework
    - Action taken to bring rejected items to compliance with product requirements, specifications, and stakeholder expectations
    - Can be expensive, hence avoid this need by good quality planning and quality assurance
  - Process adjustments
    - Correct or prevent further quality problems
    - Often results in updates to organization process assets and project management plan

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Tools and Techniques for Quality Control

- Seven basic tools of quality that help in performing quality control
  - Cause-and-effect diagrams
  - Control chart
  - Checksheet
  - Scatter diagram
  - Histogram
  - Pareto chart
  - Stratification (alternatively, flow charts or run charts)
- Statistical sampling
- Six sigma
- Testing

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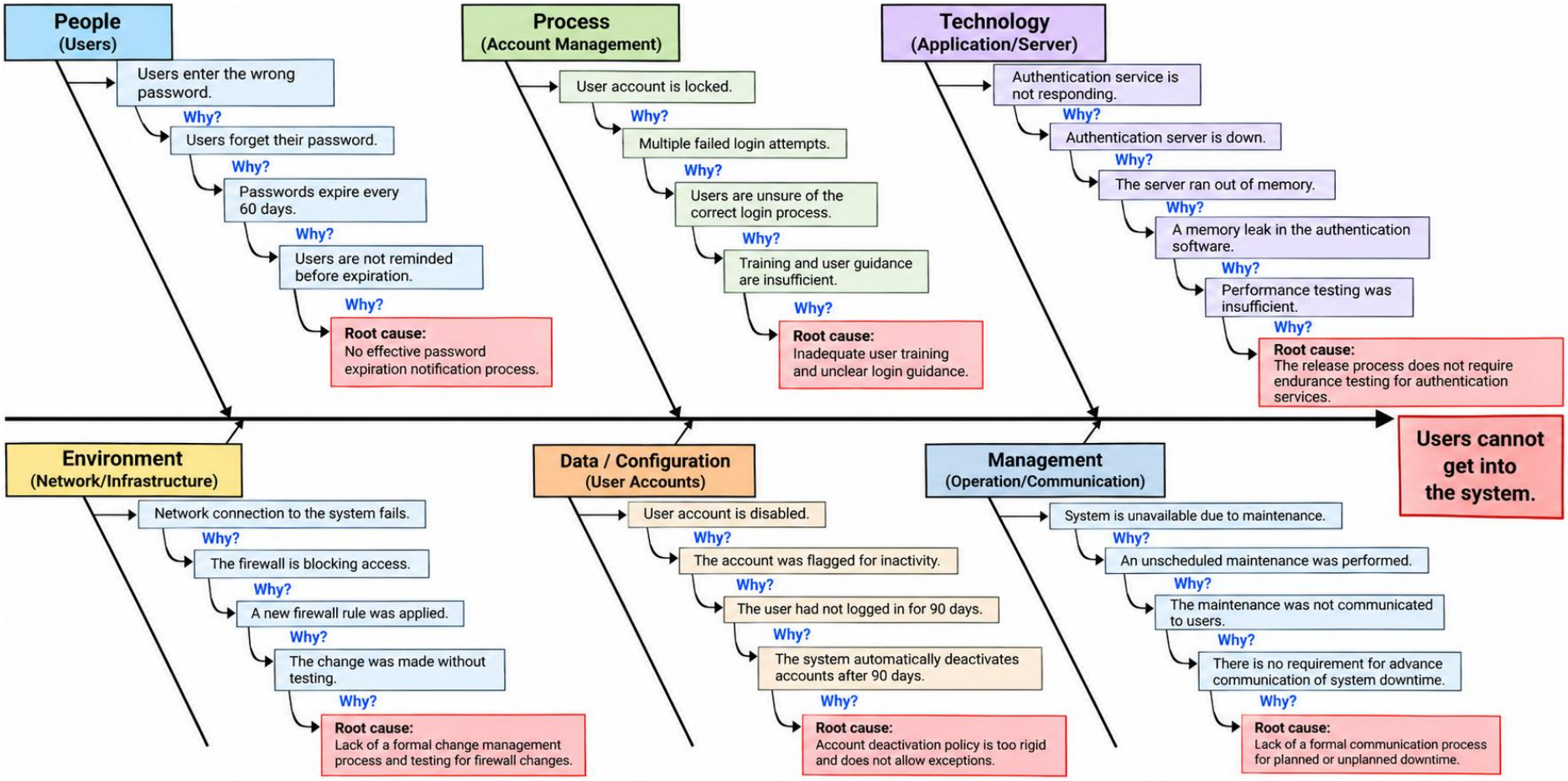
Seven Basic Tools of Quality (1 of 22)

- Cause and effect diagram (Fishbone or Ishikawa diagram)
  - Created by Kaoru Ishikawa as a cause exploration tool
  - Can use 5 whys technique to peel away the symptoms and lead to the root cause (other number can be used though)
    - The head of the fish → the problem (effect) you want to explain.
    - The bones → main cause categories (e.g. People-Methods-Machines-Materials-Environment-Measurement, People-Software-Infrastructure-Process).
    - The smaller bones branching out → underlying causes (intermediate and root causes).
    - Root cause → The deepest, most actionable cause (you cannot further ask "why?")
  - For readability, use a separate table for 5 whys analysis to trace selected causes that require deeper investigation to their root causes.

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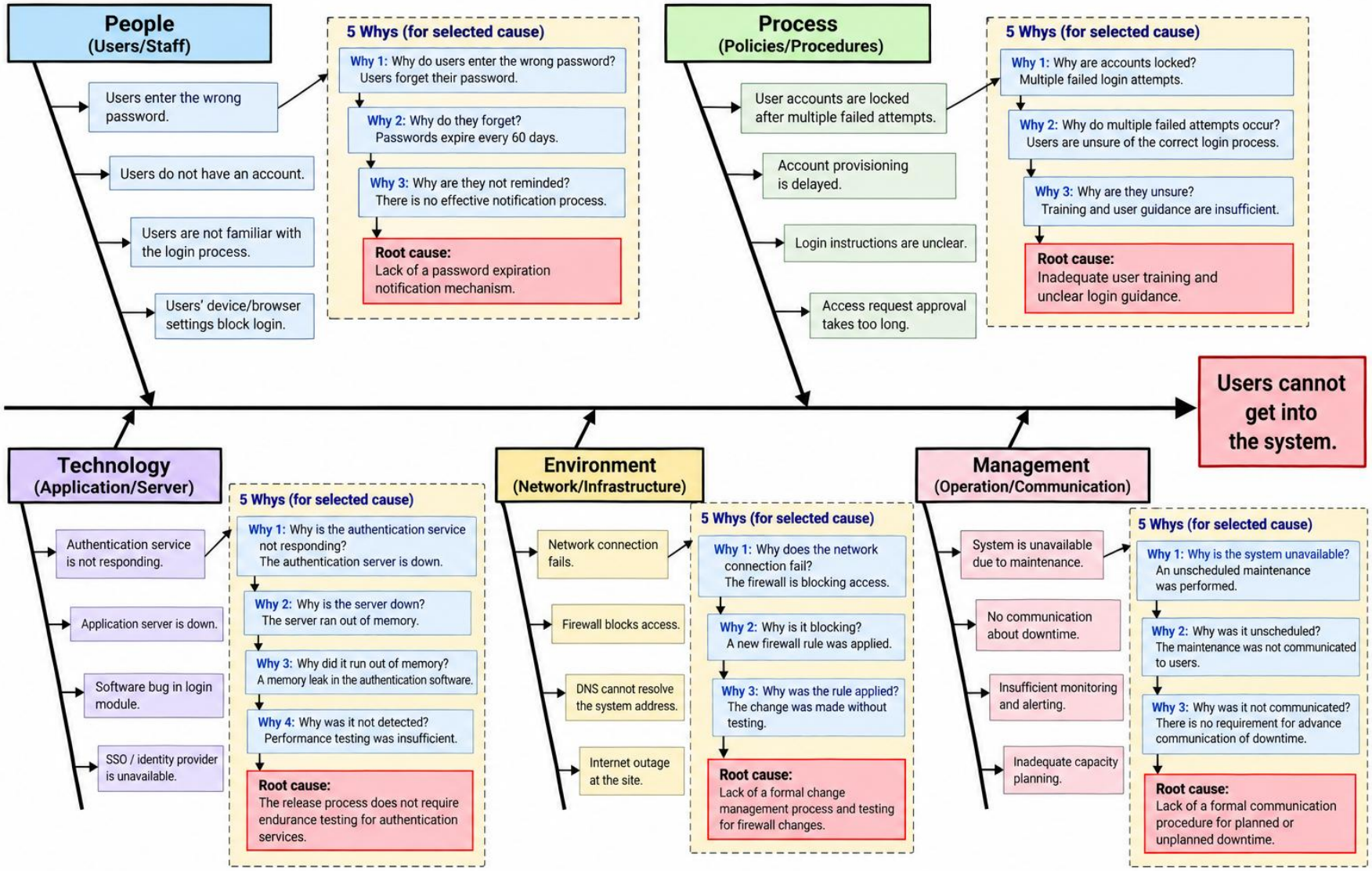
- Cause and effect diagram (cont.)



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Seven Basic Tools of Quality (3 of 22)

- Cause and effect diagram (cont.)



Seven Basic Tools of Quality (4 of 22)

- Control chart
  - Illustrates the results of the process over time, i.e. how stable the process is and whether the process creates defective products, requiring corrective action
  - Control limits (Lower Control Limit (LCL) and Upper Control Limit (UCL)) are calculated from process performance, and are used to assess whether the process is stable and operating predictably.
    - When process is in control, variations in the results are created by random events, and no need to adjust the process.
      - Data points are within LCL and UCL.
    - When process is out of control, variations in the results are created by nonrandom events, and cause of the events are to be identified and the process adjusted.
      - Data points are outside LCL or UCL.
      - Seven Run occurs.
        - Seven consecutive data points are all below or all above the mean.
        - Seven consecutive data points are all increasing or decreasing.
  - Specification limits (Lower Specification Limit (LSL) and Upper Specification Limit (USL)) are defined by customers and are useful to see how capable the process is to meet customer requirements.
    - When process is in specification, the results meet customer requirements.
      - Data points are within LSL and USL.
    - When process is out of specification, the results are defective.
      - Data points are outside LSL or USL.

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Seven Basic Tools of Quality (5 of 22)

- Control chart (cont.)

Example: Applying Control Charts to Software Product Quality Control

Consider a software development project building an **online hotel reservation system**. The project team uses a control chart to monitor the response time of the hotel's room-search API.

The objective is to determine whether the software consistently meets its performance requirements and whether changes in performance indicate problems that require corrective action.

1. Define specification limits

Suppose the performance requirement is:

> The room-search API should respond in 500 milliseconds, with an acceptable range of 400–600 milliseconds under a standardized workload.

Parameter	Value	Meaning
Target	500 ms	Desired response time
Lower Specification Limit (LSL)	400 ms	Minimum acceptable value for this exercise
Upper Specification Limit (USL)	600 ms	Maximum acceptable value

Important: In actual software performance testing, a lower response time is normally better. A 350 ms response is not a quality defect. The artificial LSL of 400 ms is introduced here to demonstrate both upper and lower specification violations. It could alternatively represent an agreed two-sided performance envelope for a benchmark test.

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- Control chart (cont.)

2. Phase I — Establishing control limits

Step 1. Collect the initial 30 data points

The team conducts 30 consecutive benchmark tests.

Observations 1–10	Observations 11–20	Observations 21–30
490, 505, 498, 512, 487, 503, 495, 509, 492, 501	496, 506, 489, 511, 650, 494, 502, 497, 508, 491	504, 499, 507, 493, 510, 488, 500, 495, 505, 350

Step 2. Compute initial control limits

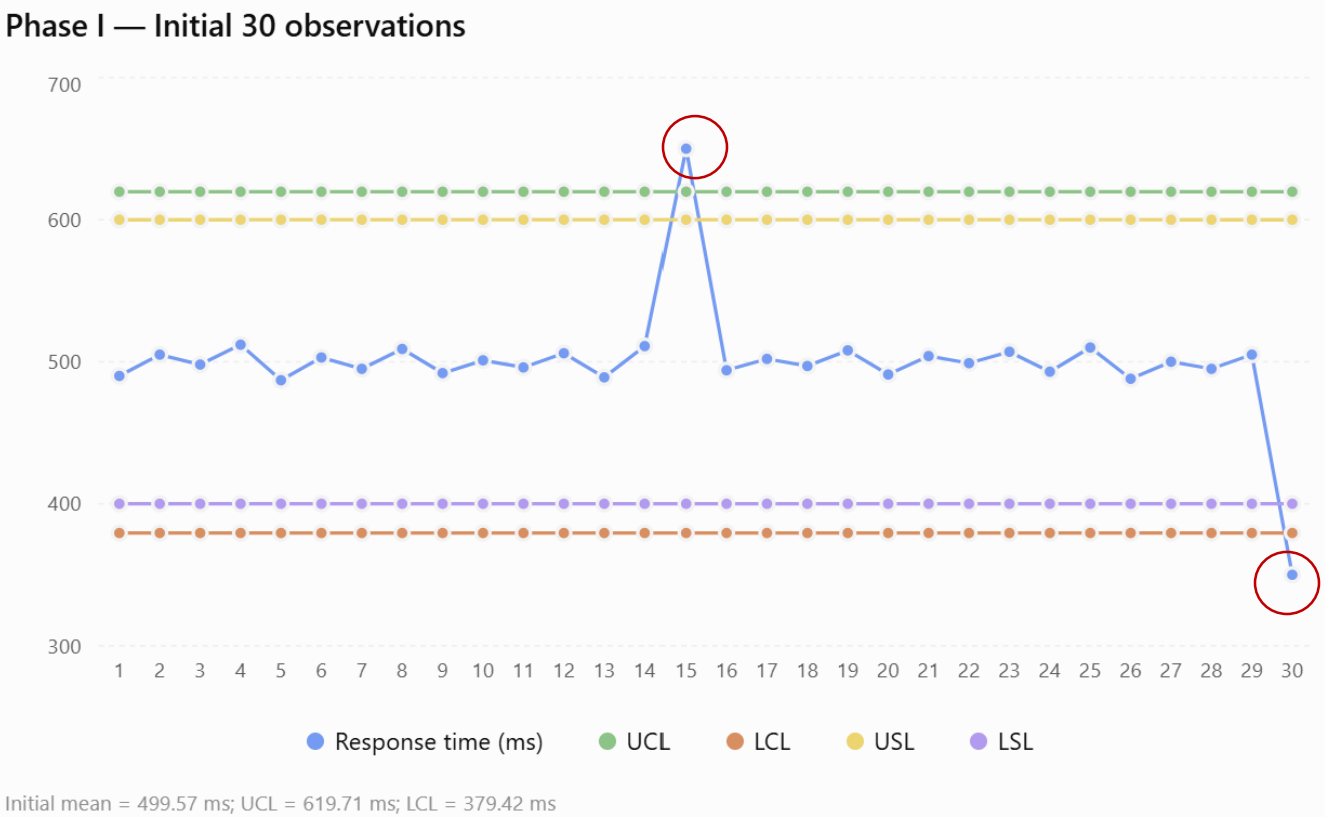
$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

$$UCL = \bar{x} + 3s$$

$$LCL = \bar{x} - 3s$$

Statistic	Value
Mean	499.57 ms
Sample SD	40.05 ms
UCL	619.71 ms
LCL	379.42 ms



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Seven Basic Tools of Quality (7 of 22)

- Control chart (cont.)

Step 3. Investigate and clean abnormal observations

Suppose investigation reveals:

- **Observation 15 (650 ms):** A backup job accidentally ran during the benchmark, creating abnormal server load.
- **Observation 30 (350 ms):** The test bypassed the database because a mock service was accidentally enabled.

These are assignable causes. The team corrects the test configuration and excludes the two invalid baseline observations, documenting the reasons.

Outliers should not be removed merely because they fall outside the control limits. If an extreme observation reflects the actual software process, it is evidence of a problem, not data to discard.

Step 4. Recompute control limits

Using the remaining 28 observations:

Mean	Sample SD	Baseline observations
499.54	**7.49**	**28**
ms	ms	valid points

The revised limits are:

$$UCL = 499.54 + 3(7.4857) = 521.99$$

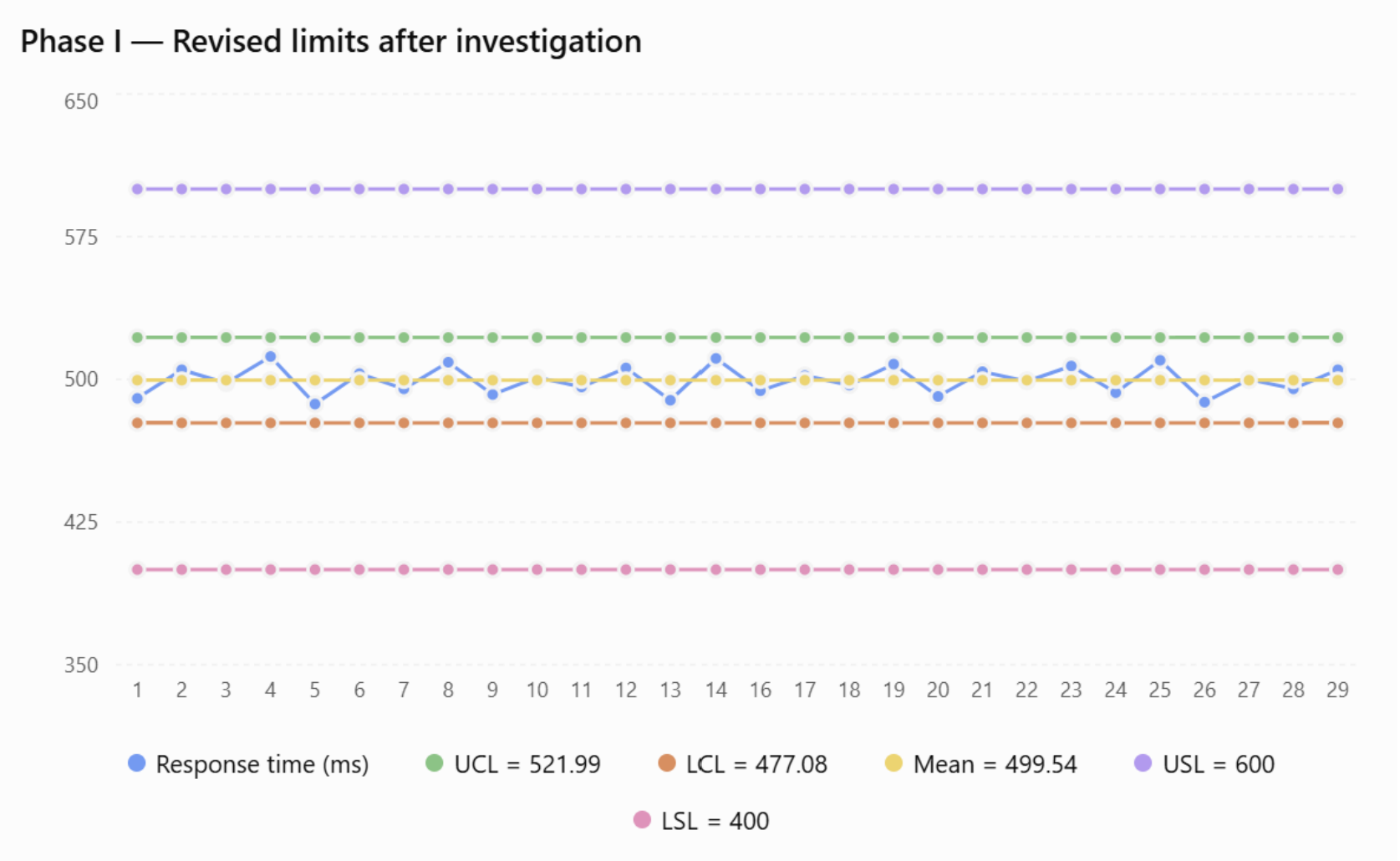
$$LCL = 499.54 - 3(7.4857) = 477.08$$

The control limits are now substantially narrower because the abnormal observations no longer inflate the estimate of process variation.

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Seven Basic Tools of Quality (8 of 22)

- Control chart (cont.)



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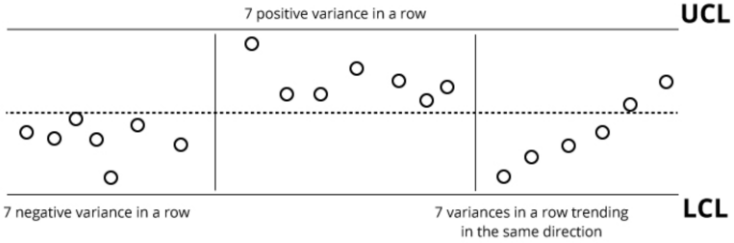
Seven Basic Tools of Quality (9 of 22)

- Control chart (cont.)

4. Phase II — Monitoring ongoing performance

Step 1. Add new data points and plot against the established limits

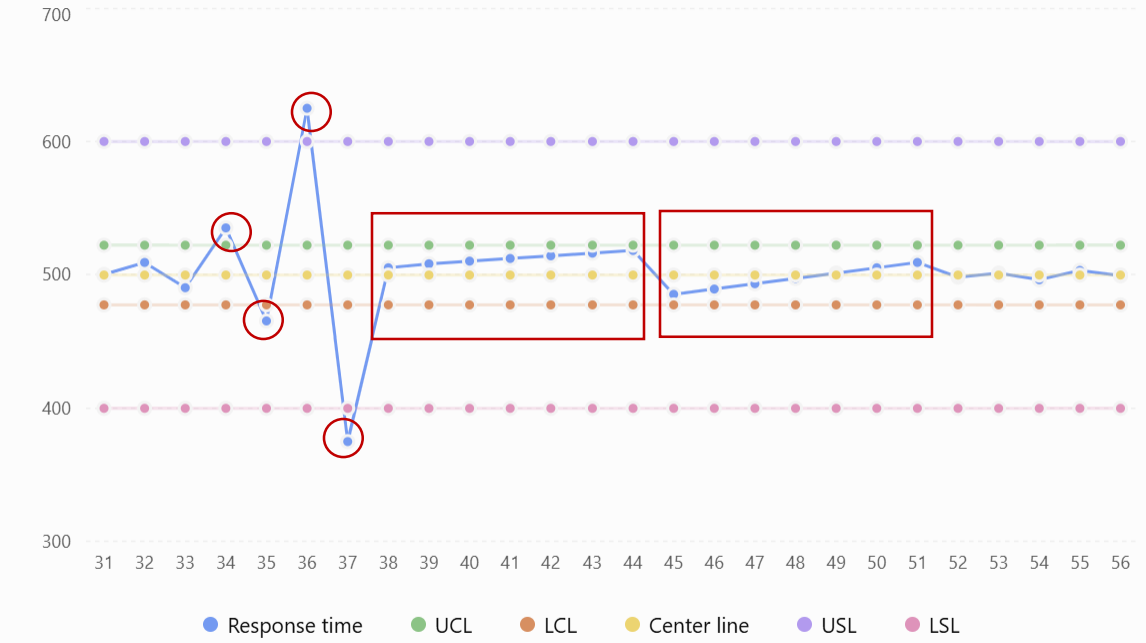
After the baseline has been established, the team performs 26 additional benchmark tests, observations 31- 56.



https://www.knowledgehut.com/tutorials/project-management/the-rule-of-seven

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Phase II monitoring chart



The chart uses the original Phase I control limits. This is essential for detecting changes rather than allowing new abnormal observations to redefine what is normal.

Examples of the six requested cases

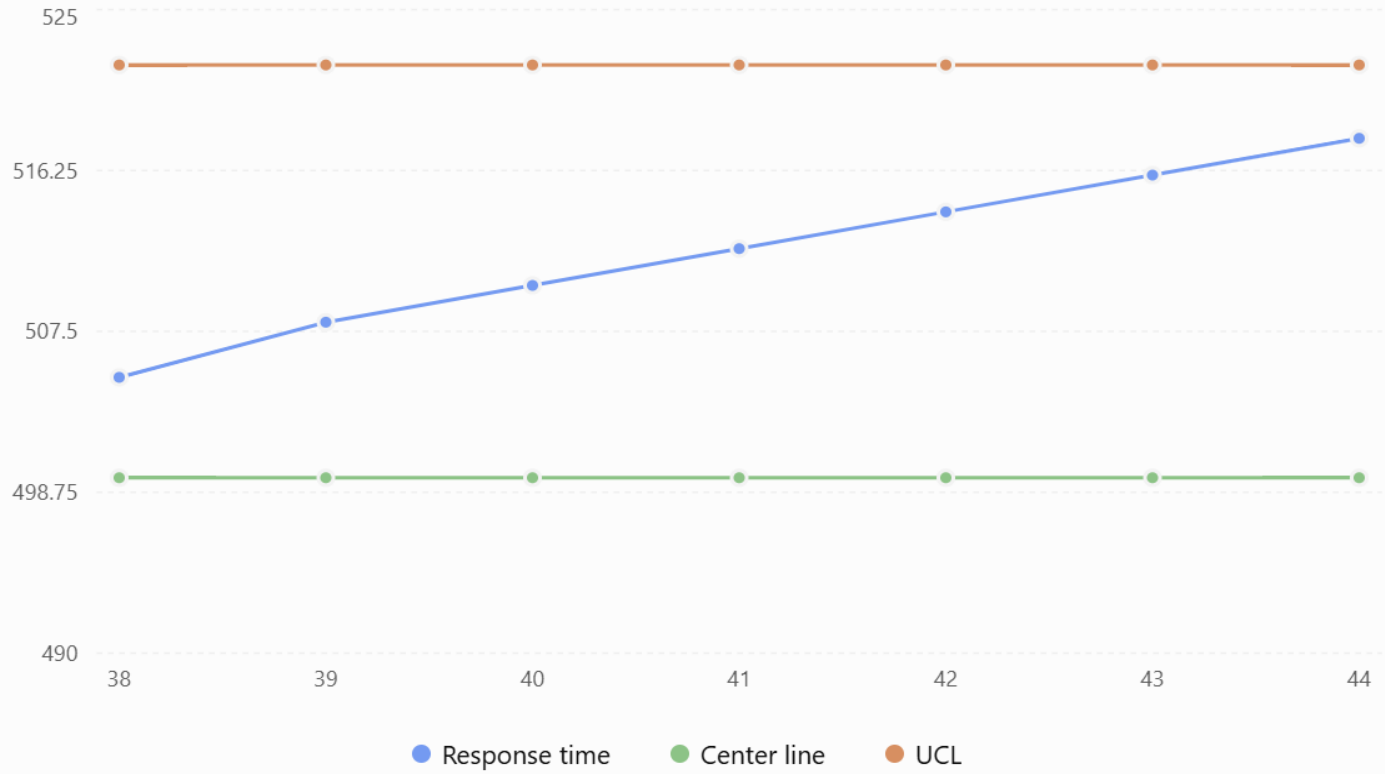
Case	Observation	Response time	Interpretation
Above USL	36	625 ms	Out of spec and out of control
Above UCL, below USL	34	535 ms	In spec but out of control
Below LSL	37	375 ms	Out of spec and out of control
Below LCL, above LSL	35	465 ms	In spec but out of control
Within control limits	31	500 ms	In control by the individual-point rule
Within specification limits	32	509 ms	In spec and within control limits

Seven Basic Tools of Quality (10 of 22)

- Control chart (cont.)

Seven-point run

Observations 38–44 are all above the center line, although each remains within the control limits.



Interpretation: Seven consecutive points above the mean suggest that response times have shifted upward. A possible cause is a recent code deployment that introduced additional database queries.

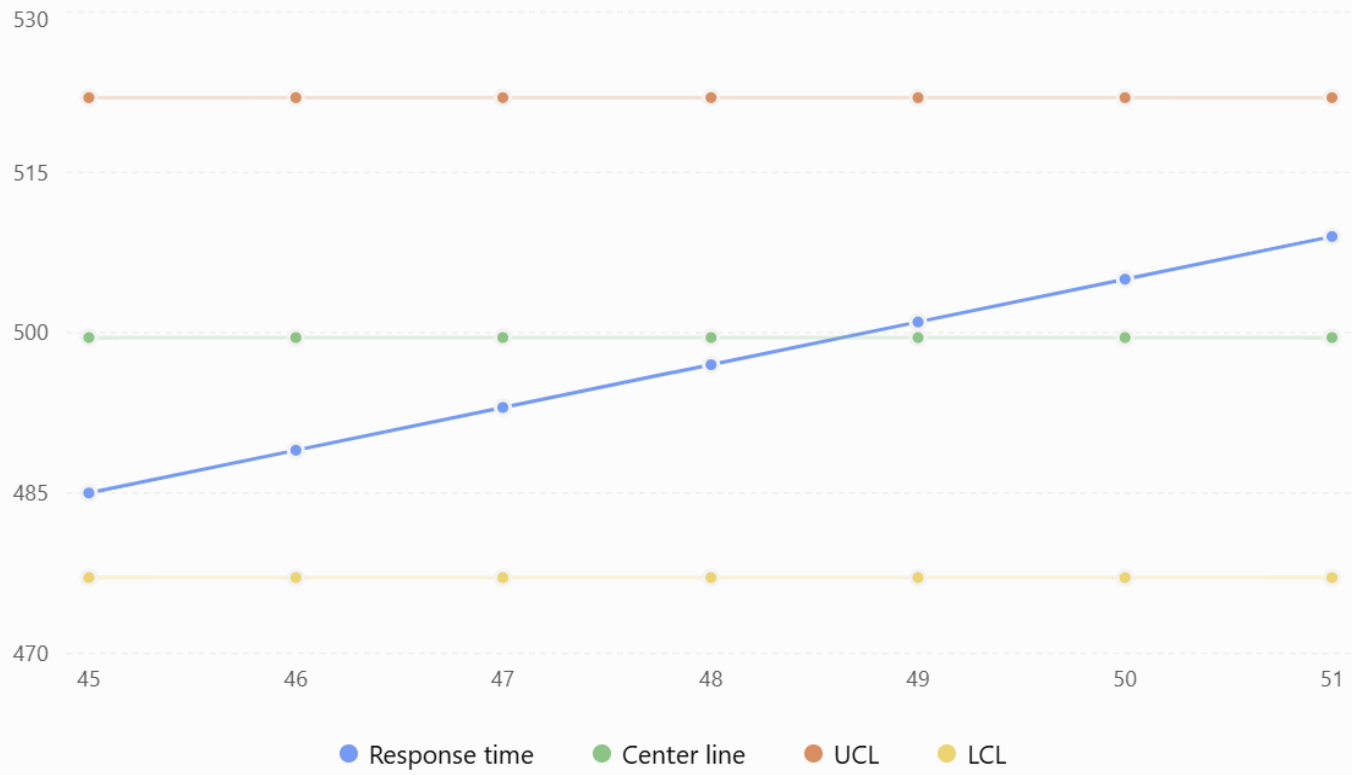
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Seven Basic Tools of Quality (11 of 22)

- Control chart (cont.)

Seven-point upward trend

Observations 45–51 increase consecutively, although all remain within the control limits.



Interpretation: The consistent increase may indicate a progressive performance problem, such as a memory leak, growing database size, or resource contention.

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- Control chart (cont.)

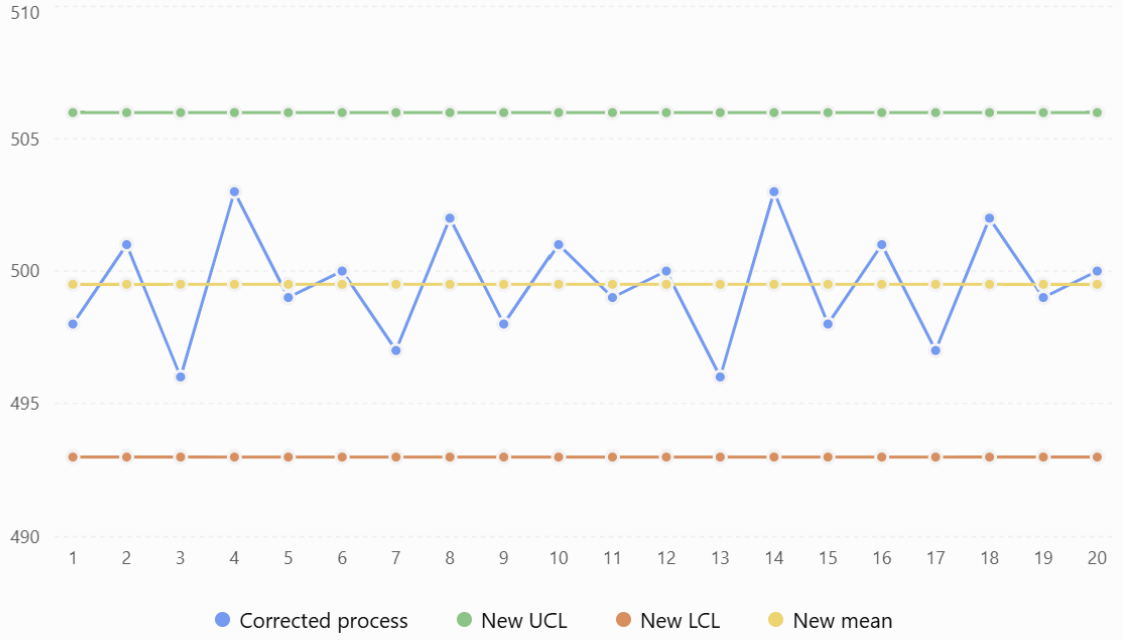
6. Phase II — Step 2. Recompute control limits when the process changes

Suppose the team identifies and fixes the causes of abnormal performance by optimizing SQL queries, correcting the test environment, and resolving a memory leak.

The team then collects 20 new observations under the corrected process:

498, 501, 496, 503, 499, 500, 497, 502, 498, 501, 499, 500, 496, 503, 498, 501, 497, 502, 499, 500

Statistic	Original stable process	Corrected process
Mean	499.54 ms	499.50 ms
SD	7.49 ms	2.16 ms
LCL	477.08 ms	493.01 ms
UCL	521.99 ms	505.99 ms
LSL	400 ms	400 ms
USL	600 ms	600 ms



The new process exhibits much less variation. The standard deviation decreased from 7.49 ms to 2.16 ms, approximately a **71% reduction**.

This suggests that the corrected process is more consistent. After confirming that the improvement is sustained and the process is stable, the new control limits can be adopted for future monitoring. Ideally, more baseline observations should be collected before finalizing the new limits.

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Seven Basic Tools of Quality (13 of 22)

- Control chart (cont.)

Statistic	Original stable process	Corrected process
Mean	499.54 ms	499.50 ms
SD	7.49 ms	2.16 ms
LCL	477.08 ms	493.01 ms
UCL	521.99 ms	505.99 ms
LSL	400 ms	400 ms
USL	600 ms	600 ms

7. Which process conditions are good, and which require correction?

Process condition	Example	Assessment	Recommended action
In spec, in control	500 ms	Good	Continue monitoring
In spec, out of control	535 ms	Needs investigation	Identify special cause
Out of spec, out of control	625 ms	Unacceptable	Correct performance problem
Out of spec, out of control	375 ms	Outside illustrative envelope	Investigate the lower-bound violation
Seven-point run	505–518 ms	Warning	Investigate process shift
Seven-point trend	485–509 ms	Warning	Investigate progressive deterioration

An additional important possibility is **a process that is in control but out of specification**. For example, a process consistently producing response times around 630 ms with narrow variation might be statistically stable, yet consistently violate the 600 ms USL. Such a process requires redesign or improvement, not merely special-cause investigation.

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Seven Basic Tools of Quality (14 of 22)

- Checksheet
  - Tally sheet or checklist
  - Used to collect and analyze data

Most complaints arrive via text messages and on Monday and Tuesday.

The information might be useful in improving the process for handling complaints.

System Complaints

Source	Day							
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Total
E-mail	𝍸	𝍸	𝍸	𝍸	𝍸		𝍸	12
Text	𝍸 𝍸	𝍸	𝍸	𝍸	𝍸	𝍸	𝍸	29
Phone call	𝍸	𝍸	𝍸	𝍸	𝍸	𝍸		8
Total	11	10	8	6	7	3	4	49

FIGURE 8-4 Sample checksheet

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Seven Basic Tools of Quality (15 of 22)

- Scatter diagram
  - Helps to show if there is a relationship between two variables
  - The closer data points are to a diagonal line, the more closely the two variables are related

Older users are less satisfied with the system.

Improvement decision can be based on the finding, e.g. system might be improved for older users.



FIGURE 8-5 Sample scatter diagram

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Seven Basic Tools of Quality (16 of 22)

- Histogram
  - Bar graph of a distribution of variables
  - Each bar represents an attribute or characteristic of a problem or situation, and the height represents frequency



Sample histogram

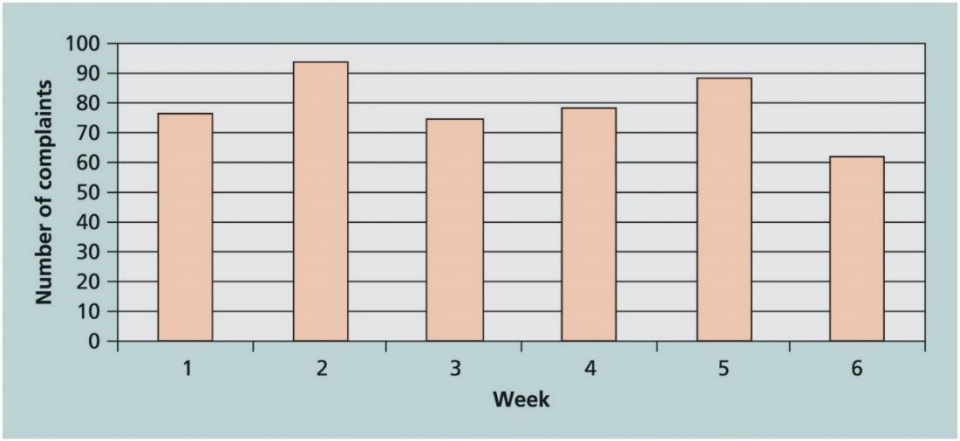


FIGURE 8-6 Sample bar graph

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Seven Basic Tools of Quality (17 of 22)

- Pareto chart
  - Contains both bars and a line graph
  - Individual values are represented in descending order by bars
  - Cumulative total is represented by the line
  - Named after Vilfredo Pareto, a noted Italian economist
    - Pareto principle or 80/20 rule: roughly 80% of the effects come from 20% of the causes

The first two complaints (90+50) account for almost 80% (77.77%) of the total complaints (180).

The company should
- Focus on making it easier to log in and addressing why the system locks up
- Find out if complaints about system being too slow are due to login or lockup problems
- Identify first who made complaints about inaccurate reports before spending more efforts

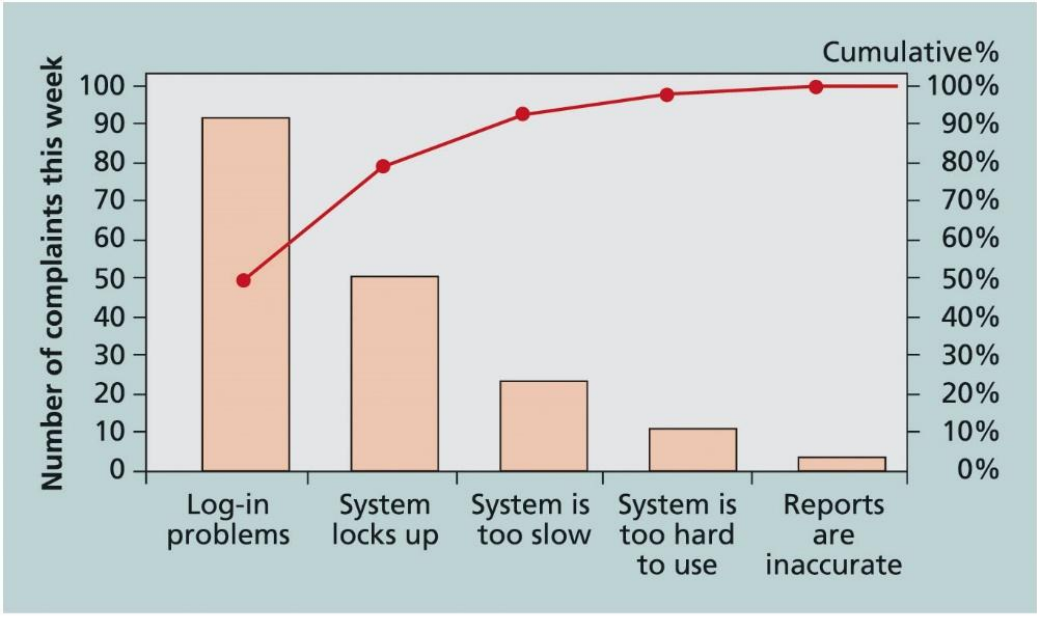


FIGURE 8-7 Sample Pareto chart

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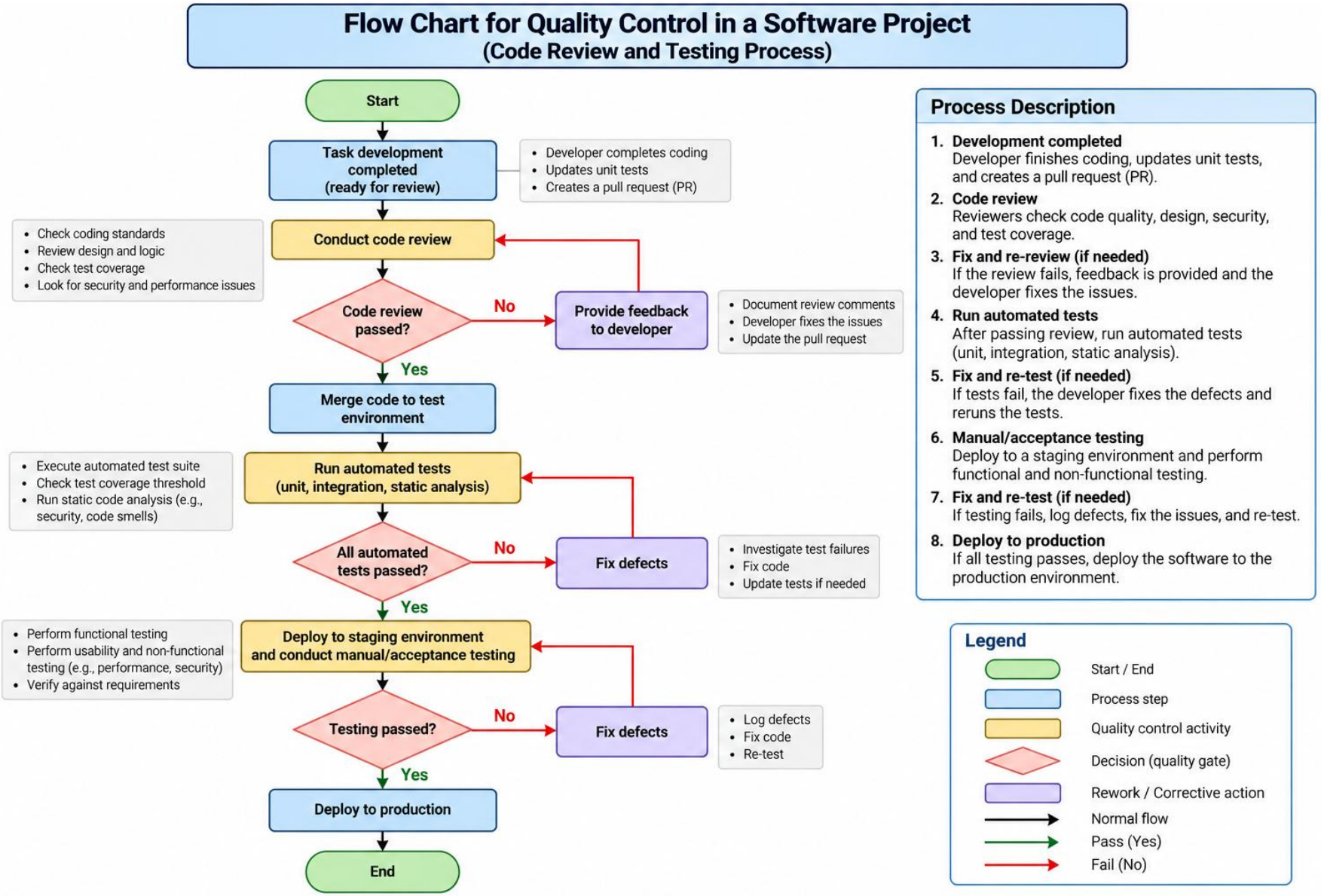
Seven Basic Tools of Quality (18 of 22)

- Flowchart
  - Visualizing the logic and flow of processes to help you understand how activities and decisions are connected.
  - Helping to identify potential bottlenecks and rework loops.
  - Providing a basis for collecting metrics and improving the process.

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Seven Basic Tools of Quality (19 of 22)

- Flowchart (cont.)



Seven Basic Tools of Quality (20 of 22)

- Flowchart (cont.)

2. Example of Quality Control Criteria

Each decision point in the flowchart represents a **quality gate**. The team should establish measurable acceptance criteria for these gates.

Quality control activity	Example acceptance criteria	Action if failed
Code review	No unresolved critical review findings	Fix code and repeat review
Unit testing	100% of required tests pass	Fix defects and rerun tests
Code coverage	At least 80% statement coverage	Add tests
Static code analysis	No critical security vulnerabilities	Correct vulnerabilities
Functional testing	All critical test cases pass	Fix defects and retest
Performance testing	95th percentile response time ≤ 2 seconds under specified load	Investigate and optimize
User acceptance testing	All mandatory acceptance criteria satisfied	Correct outstanding issues

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Seven Basic Tools of Quality (21 of 22)

- Flowchart (cont.)

3. How the Flowchart Supports Quality Control

A flowchart is one of the seven basic quality tools commonly discussed in project quality management. It helps the project team understand how activities, decisions, and rework are connected.

For example, suppose 20 pull requests enter the code review process:

- 12 pass the first review.
- 8 require corrections.
- Of those 8, 6 pass the second review.
- The remaining 2 require a third review.

The team can use the flowchart to identify where rework occurs and measure process performance.

One useful metric is the **First-Pass Yield (FPY)**:

$$FPY = \frac{\text{Items passing the first review}}{\text{Total items reviewed}} \times 100$$

$$FPY = \frac{12}{20} \times 100 = 60\%$$

A 60% first-pass yield means that 40% of pull requests required at least one round of rework. This does not automatically mean quality is poor, but it gives the team a measurable starting point for investigation.

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Seven Basic Tools of Quality (22 of 22)

- Run chart
  - Line chart that shows data points in the order of occurrence
  - Displays the history and pattern of variation of a process over time
  - Used to perform trend analysis and forecast future outcomes based on historical results

Patterns
- Defect 1 increasing over time
- Defect 2 decreasing for several months before holding steady
- Defect 3 fluctuating each month

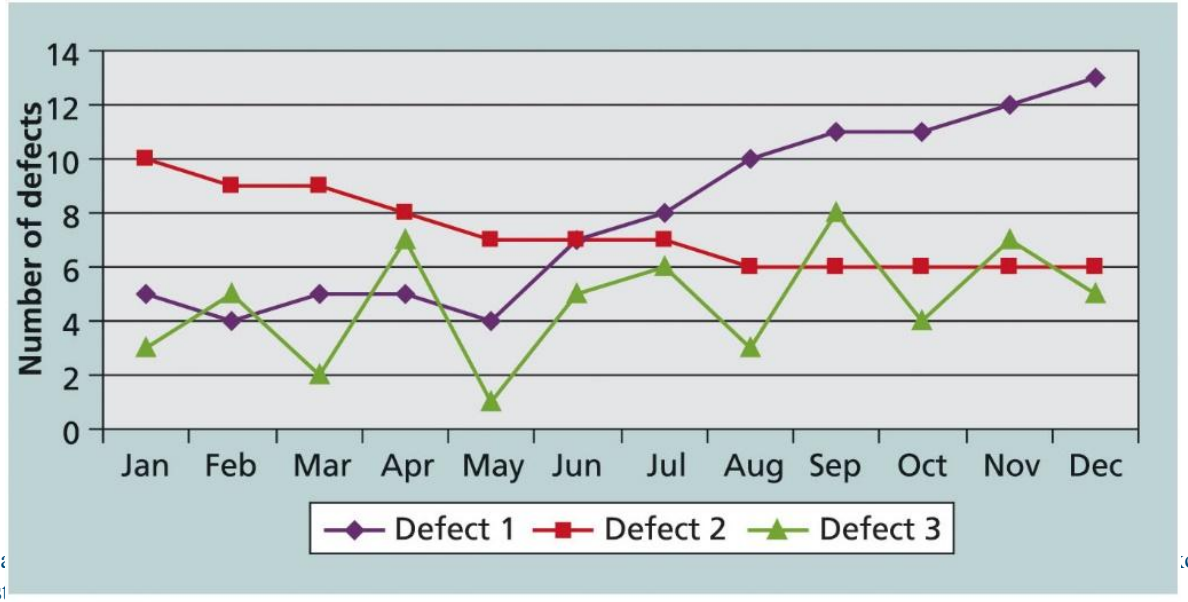


FIGURE 8-9 Sample run chart

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Statistical Sampling

- Choosing part of a population of interest for inspection
  - Size of a sample depends on how representative you want the sample to be of the population
  - Sample size formula (Cochran's formula for unknown or large population)

$$Sample\ size = 0.25 * \left(\frac{certainty\ \ factor}{acceptable\ error} \right)^2$$

    - Certainty factor = how confident you want to be that there is no variation that is not included in the samples (i.e. Z score of confidence level)
    - Acceptable error = how much error to allow (i.e. margin of error or significance level or α) = 1 - (percent certainty/100)

Desired Certainty (confidence level)	Certainty Factor (Z score)	Acceptable Error	Sample Size
99%	2.576	0.01	16,589
95%	1.960	0.05	384
90%	1.645	0.1	68
80%	1.281	0.2	10

https://www.statisticshowto.com/probability-and-statistics/find-sample-size/

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Six Sigma (1 of 8)

- Pioneered at Motorolla in 1980's and then at General Electric
- Data-driven quality management methodology used to reduce defects, minimize process variation, and improve process performance.

6. Six Sigma vs. traditional software quality control

Aspect	Traditional quality control	Six Sigma
Primary focus	Detect and correct defects	Reduce defects and process variation
Approach	Testing, inspection, reviews	Data-driven process improvement
Main question	Does the software meet requirements?	Why do defects occur, and how can they be prevented?
Typical tools	Test cases, checklists, inspections	DMAIC, Pareto charts, fishbone diagrams, statistical analysis, control charts
Success measure	Defects found and corrected	Sustained reduction in defects and variation

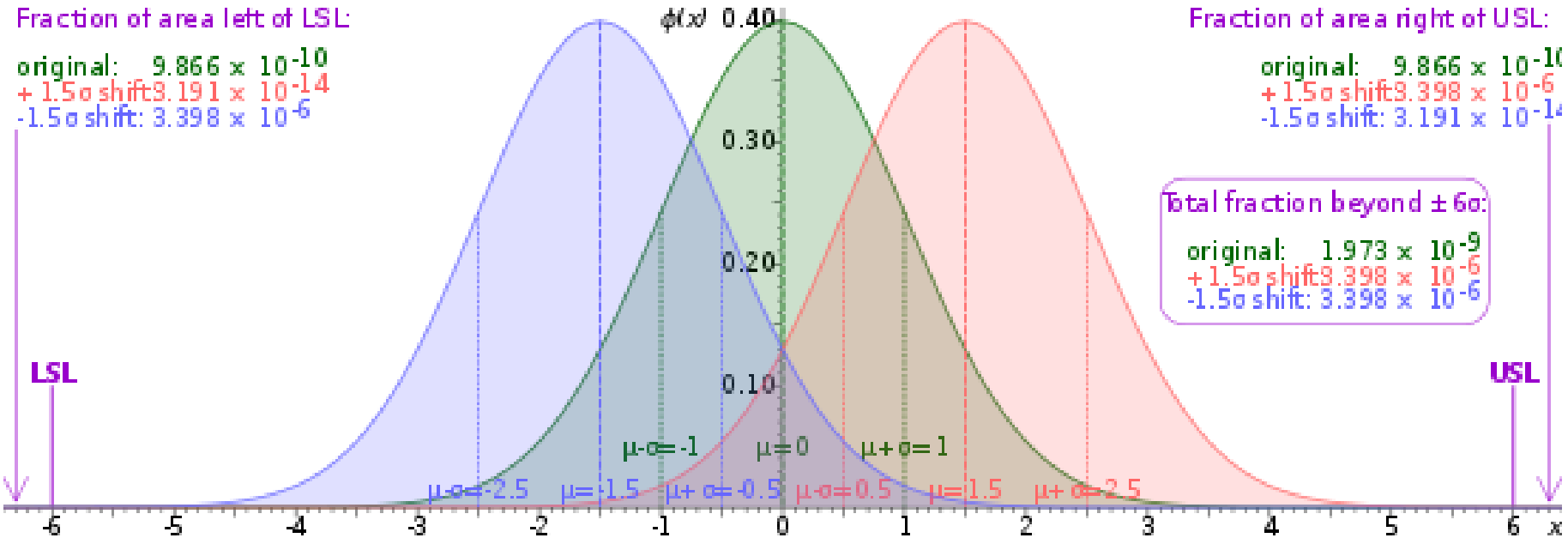
The two approaches are complementary. Software testing detects failures, while Six Sigma provides a structured method for understanding their causes and improving the development process.

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Six Sigma (2 of 8)

- The term Six Sigma itself comes from standard deviation (σ)
  - Standard deviation measures how much variation exists in a distribution of data;
  - Six Sigma projects strive for no more than 3.4 defects per million opportunities
  - However, most organizations today follows the methodology, rather than the defect limit

Over a long time, a six sigma process is allowed to shift 1.5σ on both sides from the mean due to poor control or technology.



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https://en.wikipedia.org/wiki/Six_Sigma

Six Sigma (3 of 8)

5. Six Sigma levels and software quality

The following table shows conventional Six Sigma performance levels under the 1.5σ shift convention.

Sigma level	DPMO (approx.)	Defect-free opportunities
1σ	691,462	30.85%
2σ	308,538	69.15%
3σ	66,807	93.32%
4σ	6,210	99.379%
5σ	233	99.9767%
6σ	**3.4**	**99.99966%**

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Six Sigma (4 of 8)

In Six Sigma, the **number of defect opportunities** refers to the total number of chances for a defect to occur during a process.

An **opportunity** is a specific requirement or quality characteristic that could fail. One software transaction or test case may have multiple defect opportunities.

What counts as one opportunity?

An important rule is that **defect opportunities must be clearly defined and consistently counted**.

For example, a login function might have these opportunities:

1. Correctly validate the username.
2. Correctly validate the password.
3. Correctly enforce account-lockout rules.
4. Correctly create an authenticated session.

Thus, one login attempt has four defined defect opportunities.

However, a project could instead define the entire login attempt as one opportunity, with the quality requirement being successful and correct login processing.

Both approaches can be valid, but they produce different DPMO values.

Key distinction: A defect opportunity is not necessarily a test case, a line of code, or a software function. It is a predefined chance for a particular quality requirement to be violated.

For meaningful Six Sigma comparisons, the team must use the same opportunity definitions across releases or projects.

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Six Sigma (5 of 8)

Example: Online shopping checkout

Suppose an online shopping system has five quality requirements for each checkout transaction:

No.	Defect opportunity	Example of a defect
1	Validate customer information	Accepts an invalid delivery address
2	Calculate total price	Calculates an incorrect total
3	Apply discount	Applies an incorrect discount
4	Process payment	Charges the wrong amount
5	Create order	Fails to save the order

Each checkout transaction has **5 defect opportunities**.

Suppose the team tests 1,000 checkout transactions.

$$\text{Total Opportunities} = 1,000 \times 5 = 5,000$$

During testing, the team discovers 25 defects across these opportunities.

The Defects Per Million Opportunities (DPMO) is:

$$DPMO = \frac{25}{5,000} \times 1,000,000$$

$$\boxed{DPMO = 5,000}$$

This means that the software process produces an estimated 5,000 defects per million opportunities.

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Six Sigma (6 of 8)

For example, consider a mature payment-processing service handling 10 million transactions.

Suppose its defect opportunity is defined as: *A valid payment request must be processed correctly.*

If only 20 requests fail because of software defects:

$$DPMO = \frac{20}{10,000,000} \times 10^6$$

$$\boxed{DPMO = 2}$$

This is better than the conventional Six Sigma benchmark of 3.4 DPMO.

However, this result applies only to the defined payment-processing operation. It does not establish that the entire software system is Six Sigma quality.

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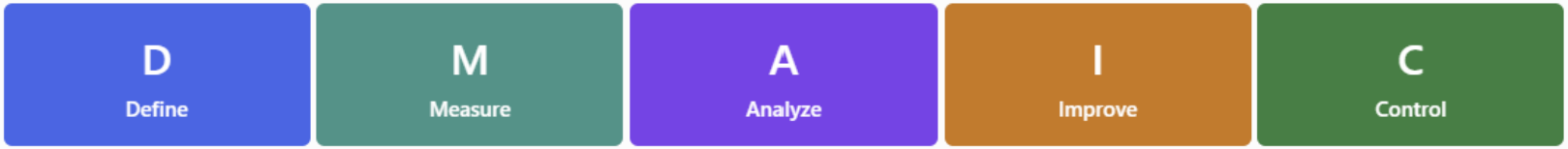
Six Sigma (7 of 8)

- **Six Sigma performance level:** A statistical quality benchmark associated with 3.4 DPMO.
- **Six Sigma methodology:** The DMAIC approach for systematically measuring, analyzing, reducing, and controlling defects.

A software project can benefit substantially from the second without attaining the first.

3. Six Sigma uses the DMAIC methodology

DMAIC is a structured, five-phase approach for improving an existing process.

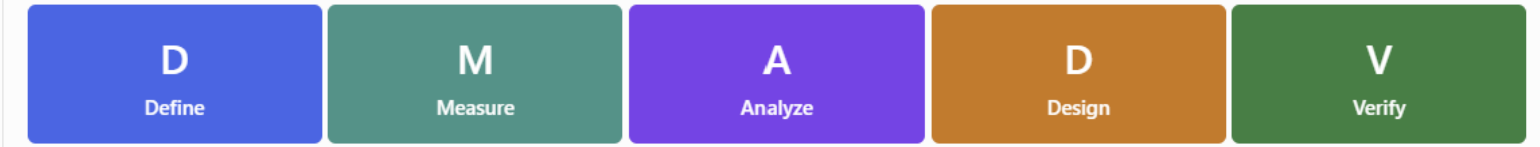


Phase	Purpose	Example in software development
Define	Identify the quality problem and improvement goal	Too many bugs escape to production
Measure	Collect data to quantify current performance	Measure production defects per 1,000 test opportunities
Analyze	Identify root causes of defects	Investigate inadequate testing and unclear requirements
Improve	Implement solutions to remove root causes	Introduce automated testing and code reviews
Control	Monitor performance to sustain improvements	Use control charts and regular quality audits

Six Sigma (8 of 8)

DMADV is a Six Sigma methodology used to design new products, services, or processes that meet customer requirements and quality expectations from the beginning.

1. Five phases of DMADV

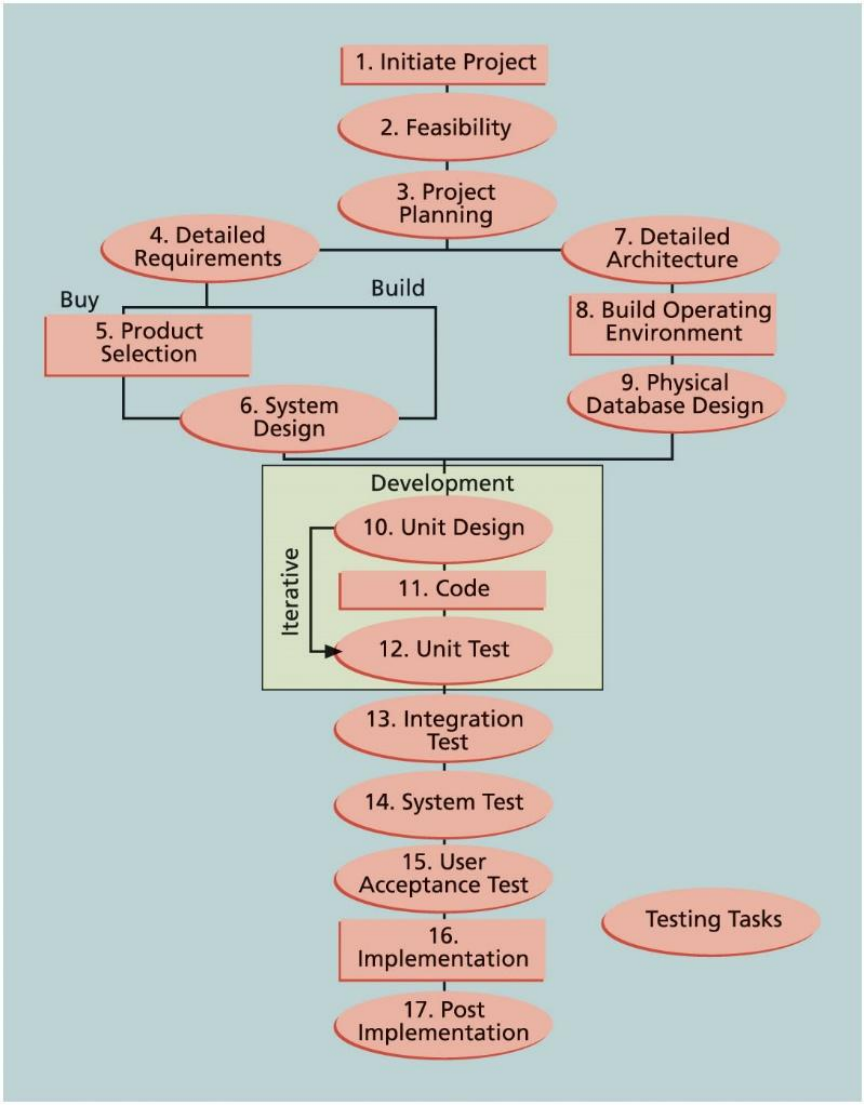


Phase	Purpose	Application in software projects
Define	Define customer needs and project objectives	Identify the quality goals of a new software system
Measure	Translate customer needs into measurable requirements	Define response time, reliability, and defect targets
Analyze	Evaluate alternative designs and identify risks	Compare architectures and technical solutions
Design	Develop the selected solution	Design architecture, database, interfaces, and software components
Verify	Verify that the design meets requirements	Conduct testing, performance evaluation, and user acceptance testing

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Testing (1 of 4)

- Many IT professionals think of testing as a stage that comes near the end of IT product development
- Testing tasks need to be done during almost every phase of the systems development life cycle, not just before the organization ships or hands over a product to the customer



Source: Hollstadt & Associates, Inc.

FIGURE 8-11 Testing tasks in the software development life cycle

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Testing (2 of 4)

Testing and other quality evaluation activities can be divided into two broad categories:

- **Static testing:** Evaluating requirements, designs, or source code without executing the software. Examples include requirements reviews, design inspections, and static code analysis.
- **Dynamic testing:** Executing software to determine whether it behaves correctly. Examples include unit testing, integration testing, and system testing.

During the early SDLC phases, static testing is particularly important because executable software may not yet exist.

SDLC Phase	Testing Activities	Example
1. Requirements Analysis	**Requirements validation**	Check whether the requirement "The system must process orders quickly" is clear, complete, and testable.
2. System Design	**Design review and validation**	Review the database design to ensure that one customer can place multiple orders.
3. Implementation	**Unit testing**	Test whether the function calculating the total order price produces the correct result.
4. Integration	**Integration testing**	Test whether the ordering module correctly communicates with the payment module.
5. System Testing	**System testing**	Test the complete ordering process, from selecting food to making payment and receiving confirmation.
6. Deployment	**Acceptance testing and smoke testing**	Verify that the deployed system works correctly and meets business requirements.
7. Maintenance	**Regression testing**	After adding a discount feature, verify that existing price calculations still work correctly.

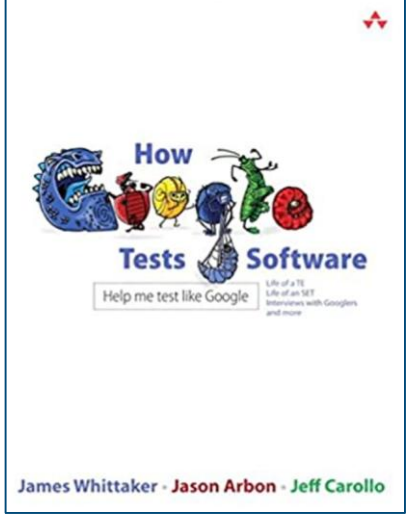
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Testing (3 of 4)

- Types of tests
  - Unit testing
    - Tests each individual component (often a program) to ensure it is as defect-free as possible
  - Integration testing
    - Occurs between unit and system testing to test functionally grouped components
  - System testing
    - Tests the entire system as one entity
  - User acceptance testing
    - Is an independent test performed by end users prior to accepting the delivered system

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Testing (4 of 4)



- Testing alone is not enough
  - Watts S. Humphrey, a renowned expert on software quality, defines a software defect as anything that must be changed before delivery of the program
- Testing does not sufficiently prevent software defects
  - The number of ways to test a complex system is huge
  - Users will continue to invent new ways to use a system that its developers never considered
- Humphrey suggests that people rethink the software development process to provide no potential defects when you enter system testing
  - Developers are responsible for the quality of their own work
  - Developers do not rely on testers to ensure quality.

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Modern Quality Management

- Modern quality management:
  - Requires customer satisfaction
  - Prefers prevention to inspection
  - Recognizes management responsibility for quality, e.g.
    - Provide training
    - Give recognition
    - Remove barriers

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Improving IT Project Quality

- Suggestions for improving quality for IT projects
  - Establish leadership that promotes quality
  - Understand the cost of quality
  - Work toward improving the organization's overall maturity level in software development and project management

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Leadership

- A large percentage of quality problems are associated with management, not technical issues
  - Top management must be quality-minded, taking responsibility for creating, supporting, and promoting quality programs
- Leadership provides work environment conducive to producing quality
  - Top management stresses the need to develop and use quality standards
  - Top management provides resources such as staff, training, customer input, dedicated workspace, and quiet work environment to improve quality

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The Cost of Quality (1 of 2)

- Cost of conformance plus the cost of nonconformance
  - Conformance means delivering products that meet requirements and fitness for use
  - Cost of nonconformance means taking responsibility for failures or not meeting quality expectations

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The Cost of Quality (2 of 2)

- Cost categories related to quality
  - Prevention cost
    - Cost of planning and executing a project, so it is error-free or within an acceptable error range
  - Appraisal cost
    - Cost of evaluating processes and their outputs to ensure quality
  - Internal failure cost
    - Cost incurred to correct an identified defect before the customer receives the product
  - External failure cost
    - Cost that relates to all errors not detected and corrected before delivery to the customer
  - Measurement and test equipment costs
    - Capital cost of equipment used to perform prevention and appraisal activities

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Maturity Models

- Frameworks for helping organizations improve their processes and systems
  - Capability Maturity Model Integration (CMMI) is a process improvement approach that provides organizations with the essential elements of effective processes
    - CMMI levels: Incomplete, Performed, Managed, Defined, Quantitatively Managed, Optimizing
  - Organizational Project Management Maturity Model (OPM3)
    - Released by PMI in December 2003
    - Addresses standards for excellence in project, program, and portfolio management best practices, and explains the capabilities necessary to achieve those best practices

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Using Software to Assist in Project Quality Management

- Software can be used to assist with tools and techniques
  - Spreadsheet and charting software helps create diagrams
  - Statistical software packages help perform statistical analysis
  - Specialized software products help manage Six Sigma projects or create quality control charts







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Considerations for Agile/Adaptive Environments

- Agile methods call for frequent quality and review steps built in throughout the project
- Agile methods focus on small batches of work, aiming to uncover inconsistencies and quality issues earlier in the project when overall costs of change are lower
- Agile methods is concerned with satisfying stakeholder needs, having testing by users at the end of each iteration
- Agile projects need an agreed upon definition of done that includes a quality criterion
- Retrospectives check on effectiveness of the quality process, looking for the root cause of issues and suggesting new approaches to improve quality

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Chapter Summary

- Quality is a serious issue
  - Project quality management includes planning quality management, performing quality assurance, and controlling quality
  - Many tools and techniques are related to project quality management
  - Many people made significant contributions to the development of modern quality management
  - There is much room for improvement in IT project quality
  - Several types of software are available to assist in project quality management

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